

Himanshu Dongre

Associate Technical Architect | AI/ML Architecture, Perception & Embedded Systems
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PROFILE

Technical architect with 12+ years in software and 7+ years building AI systems across autonomous driving, in-cabin intelligence, and embedded deployment. Leads model industrialization from feasibility and architecture through target-hardware validation; also builds open-source agent infrastructure.

CORE STRENGTHS

ML & PERCEPTION

Multimodal perception, in-cabin sensing, 2D/3D detection and tracking, model optimization

SYSTEMS & RUNTIME

C++, Python, Rust, Linux, PyTorch, ONNX, TensorRT, OpenCV

ARCHITECTURE & PLATFORMS

NVIDIA DRIVE/Jetson, Snapdragon Ride, heterogeneous compute, MLOps/DataOps, model industrialization

OPEN-SOURCE

SYSTEMS

SMRITI

Open-source coordination substrate for multi-agent work: versioned checkpoints, claims, structured tasks, drift detection, and worktrees. Public dogfooding: 129 checkpoints, 67 continuations, 87 claims. Website | GitHub

PROPOPS

Open-source research agent joining RERA, IGRS, and eCourts data through 19 workflows, government-data scrapers, entity resolution, and batch evaluation. GitHub repository

PATENT

Mobile Device Performance Improvement System Using Cloud Computing.
Indian patent application | 1922/MUM/2013

EXPERIENCE

KPIT TECHNOLOGIES | ASSOCIATE TECHNICAL ARCHITECT

AI/ML Systems | 2019 - Present | Pune, Tokyo, Bengaluru

- Architect AI systems across autonomous-driving perception, in-cabin intelligence, and model industrialization for global automotive R&D programs including BMW, Honda, Nissan, and Toyota Research/Woven; lead technical direction from feasibility through target-hardware validation.
- Defined and governed a multi-model in-cabin program spanning whole-body pose, gaze, and temporal action/skeleton forecasting on NVIDIA Jetson; drove model feasibility, ONNX conversion, C++ porting, quantization, and pipeline integration.
- Developed a reusable deployment architecture across NVIDIA DRIVE/Jetson, Qualcomm Snapdragon Ride, Ambarella, and Renesas platforms, covering custom operators, heterogeneous compute allocation, multi-model scheduling, and performance profiling.
- Enabled deep-learning workloads on severely constrained automotive ECUs by selecting suitable runtimes, implementing unsupported operators, and restructuring Python pipelines in C++; achieved multi-fold latency and memory improvements while preserving agreed model quality.
- Led implementation of adaptive-speed, driver-fatigue, and takeover-transition features for BMW; designed C/C++/Rust CAN interfaces for Toyota Research/Woven's Arene OS.
- Built perception middleware and vehicle interfaces spanning camera, LiDAR, radar, traffic-light recognition, vehicle tracking, and concurrent inference under strict compute, memory, reliability, and safety constraints.
- Lead architecture for MLOps and DataOps supporting model experimentation and validation; advise internal teams on agentic use cases, evaluation, orchestration, tool integration, state/memory, and human approval.

INDEPENDENT WORK | SOFTWARE ENGINEER

Bengaluru

- Built a neural style-transfer system for a mobile application and a healthcare billing platform.

ACCENTURE | SOFTWARE ENGINEER

Bengaluru

- Built telecom test automation and contributed to real-time rating, billing, and backend systems.

EDUCATION

M.S. COMPUTER SCIENCE | UNIVERSITY OF COLORADO BOULDER

In progress | GPA 4.0

PG PROGRAMME, ML & AI | IIIT BANGALORE

B.E. COMPUTER ENGINEERING | NAGPUR UNIVERSITY